



**Karolinska
Institutet**



Master's Programme in Health Informatics
Spring Semester 2026
Degree thesis, 30 Credits

Multimodal Agentic Collaboration for Clinical Treatment Planning: A Comparative Study of Orchestration Architectures

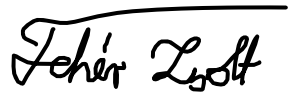
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Affirmation

I hereby affirm that this Master's thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text. This work has not been submitted for any other degree or professional qualification except as specified, nor has it been published.

Stockholm, 2026-06-08

A handwritten signature in black ink, reading "Fehér Zsolt". The signature is written in a cursive style with a horizontal line above it.

Zsolt Fehér

Abstract

Background

Agentic AI extends large language models with autonomous planning, iterative refinement, and tool use, while multi-agent systems distribute these capabilities across specialised agents that collaborate on complex tasks. In clinical treatment planning, decisions rely on integrating multimodal evidence, yet current decision support approaches lack clear comparative guidance for multi-agent design.

Aim

The aim of this project is to examine how different multi-agent architectural paradigms affect the integration, utilisation, and performance characteristics of multimodal clinical data in treatment planning tasks, with specific attention to validation and correctness.

Method

A small mixed-methods comparative study evaluated groupchat, parallel workflow, assembly chain, and nested discussion architectures on four structured cases from deidentified MIMIC-Eye and MIMIC-IV-Note data. Clinically inspired agents used local language and vision models plus retrieval from medical textbooks, PubMed, and RxNorm. Outputs were compared with withheld reference summaries using exact and partial match criteria, confidence, hallucinations, runtime, and tool use.

Results

No architecture dominated all dimensions. Groupchat and nested discussion produced the richest deliberation and the only full matches, but required longer runtimes and showed greater variability. Parallel workflow was fastest and comparatively stable, while assembly chain could propagate early unsupported assumptions. Hallucinations were mainly linked to weak grounding, timeline confusion, and unverified tool claims.

Conclusion

Multi-agent orchestration appears useful when role specialisation is paired with explicit synthesis and verification. Future systems should prioritise evidence tracking, prompt constraints, architectural guardrails, and clinician-reviewed evaluation.

Keywords:

Clinical decision support systems, large language models, multi-agent systems, multimodal clinical reasoning, orchestration architectures, retrieval-augmented generation

Acknowledgement

I would like to express my sincere gratitude to my supervisor, Professor Magnus Boman, for his guidance, encouragement, and thoughtful feedback throughout this project. His support helped me shape the research question and maintain a critical perspective on both the technical and clinical dimensions of the work.

In addition, I would like to thank the teachers and colleagues in the Master's Programme in Health Informatics for valuable discussions during the course of my studies.

Finally, I would like to thank my family and friends for their patience and support throughout this process. Their encouragement made it possible to complete this thesis.

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List of Abbreviations

AC - Assembly Chain
AI - Artificial Intelligence
BP - Blood Pressure CDSS - Clinical Decision Support System
CXR - Chest X-ray
ED - Emergency Department
EHR - Electronic Health Record
GC - Group Chat
HR - Heart Rate
ICU - Intensive Care Unit
LLM - Large Language Model
LVM - Large Visual Model
MAS - Multi-Agent System
NC - Nested Chat
PW - Parallel Workflow
RAG - Retrieval-Augmented Generation
RR - Respiratory Rate
SpO2 - Peripheral Oxygen Saturation

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1 Introduction

1.1 Background

Healthcare itself faces many challenges that require coordination among diverse specialists, each bringing unique expertise or perspective to patient care. Traditional clinical decision support systems, while valuable and well-developed, often operate as rule-based or model-specific applications that do not fully reflect the collaborative characteristics of a clinical team. Multi-agent AI systems offer an alternative by distributing roles across specialised computational agents that can communicate, negotiate, and collaborate to address complex clinical challenges [1].

The recent development of large language models (LLMs) changed the design space for clinical decision support because these models can interpret and generate clinical language, summarise patient records, and reason across long textual contexts. This made it possible to imagine decision-support tools that are less rigid than earlier systems and that can work with narrative notes, guideline text, and patient-specific context. However, LLMs also have important limitations in clinical reasoning. They can generate plausible but unsupported statements, rely on static training knowledge, fail to retrieve current or patient-specific information unless connected to tools, and show limited reliability when asked to integrate multiple sources of evidence without explicit grounding. These weaknesses are especially problematic in treatment planning, where recommendations must be traceable to the patient’s actual data and to current clinical knowledge.

Agentic AI can be seen as a response to these limitations. Instead of treating the LLM as a one-shot answer generator, agentic systems combine language models with planning, memory, tool use, retrieval, and iterative revision [3]. In a clinical setting, this allows a system to search external evidence, inspect patient-specific files, call terminology resources, and revise its conclusions after receiving new observations. The shift from single-agent to multi-agent approaches extends this idea further by distributing reasoning across specialised agents, such as a general physician, radiologist, laboratory specialist, or pharmacist. This mirrors the fact that effective healthcare decisions draw on multiple perspectives and domains of expertise [2]. Recent advances in transformers and multimodal models have also made it increasingly feasible to process diverse data types. This technological trajectory provides the motivation for the present thesis. If LLMs made flexible clinical reasoning interfaces possible, their limitations in grounding, verification, and multimodal integration explain why agentic and multi-agent designs are now relevant. The central question is therefore not only whether an LLM can produce a clinically plausible treatment plan, but whether an orchestration architecture can structure specialised reasoning, tool use, and evidence synthesis in a way that is more transparent, stable, and clinically useful.

1.1.1 Multi-agent Orchestration

Agentic AI is an emerging class of artificial intelligence systems designed to make autonomous decisions and adapt their behaviour through interaction with their environment. Building on recent advances in large language models (LLMs), these systems extend beyond conventional predictive pipelines by supporting self-directed planning, iterative refinement, and the selective use of external tools to complete tasks [3]. Multi-agent systems utilise this paradigm by combining multiple autonomous agents that collaborate with one another, enabling distributed problem-solving strategies that resemble aspects of human teamwork and social decision-making [3]. The literature defines these agentic collaboration patterns as Orchestration [4].

There is a clear distinction between these systems, as they differ in both information flow and implementation. Traditional multi-agent designs generally follow either chain-based or parallel workflows. In a chain-based workflow, agents operate in sequence, refining each other’s outputs while potentially extending the material with their own findings and perspectives [5]. By contrast, in a parallel workflow, agents are not directly connected, but instead they operate independently and are later coordinated by a summarizing agent that synthesizes the results [5]. On the other hand, with LLMs’ unique interaction-based workflows, these system designs have had an opportunity to integrate these actors in a more natural way, which the literature calls a GroupChat or a group of Conversable Agents [6]. These agents have the capability to move beyond simple output refinement, as they engage in a proper, well-bounded discussion that mimics a real-world conversation. The order of speaking can be determined by a predefined sequence or a GroupChat Manager, who may be a human or another LLM. Studies show that collaborative agent frameworks can effectively distribute specialised tasks and improve decision accuracy over single-agent models [6][7]. Notably, a cognitive bias mitigation AI system achieved a 76.6% diagnostic accuracy gain, outlining the value of structured peer-style agent dialogues in reducing errors [8]. Similarly, a triage agentic system reduced discordance rates by 18.4%, demonstrating tangible improvements in emergency triage consistency [9]. These two examples showcase how agentic conversations can positively influence clinical outcomes.

Profession-specific roles serve a crucial function in almost every architectural composition. Each of these agents focuses on one specific expert domain, which is much more efficient than a single agent that is required to be an expert in many such domains [2]. The latter are more prone to hallucinations in the case of LLMs [2]. By distributing the knowledge spaces, we gain better accuracy and reliability, while the system reflects a professional clinical team. To create such roles, context specification is essential, since it has been shown that a predefined role for an LLM can greatly enhance its output by providing a helpful starting perspective [8]. Many studies on this topic have shown a tendency toward role specification by further dividing and conquering tasks of varying complexity [8, 9, 10]. The notion of job partitioning is not a new concept in MAS architectures. However, by introducing perceptual diversity among agents, we can ensure that the divide-and-rule truism not only splits duties but also sets a notably different starting position for individual LLMs in terms of approach. The roles are typically based on real clinical positions, such as physiologist, radiologist, or pharmacist. They have a specific focus in a given scenario, as well as their own goals and reporting requirements. In practice, all of this is given in the LLMs’ context before receiving

any job or scenario.

Furthermore, in this context window, we can determine the formatting of such outputs, making them even more structured and thus reducing the chaotic nature of these responses [10]. On top of that, these structured outputs serve as protocols from a traditional perspective, since they enable standardised communication within and outside the system. Many modern LLM frameworks were built upon such conversation protocols [6]. However, the application of standardised communication protocols in such environments has been around for many decades [11], and with this newfound direction of multi-agent systems, the importance of revisiting the timeless literature is at an all-time high.

1.1.2 Autonomous Agents

Autonomous agents are computational entities that pursue high-level goals by iteratively observing their environment, deciding on actions, and updating their internal state without requiring step-by-step human instructions [1].

In common LLM-based systems, this autonomy is typically implemented through a loop of reasoning, tool use, and reflection, where the model decomposes a complex task into sub-goals, calls external tools such as functions, or EHR interfaces, and revises its plan based on intermediate results. This distinguishes autonomous agents from traditional, single-shot LLM prompts, which generate an answer once and do not maintain an explicit notion of progress, memory, or environment [12]. Several architectural patterns have emerged for implementing such agents. ReAct-style agents bounce between thought and action steps, using natural language reasoning to decide when to query external tools and how to interpret their outputs before proceeding. AutoGPT-like systems extend this idea to longer horizons as the given abstract goal, the agent proposes a plan, decomposes it into sub-tasks, executes them one by one, and maintains a working memory of past actions and observations that can be revisited or refined as the task evolves [13].

Another popular architecture is retrieval-augmented generation in which a large language model is paired with an external retrieval component that fetches relevant documents or data at query time and injects them into the model’s prompt, allowing responses to be grounded in up-to-date and domain-specific information rather than relying solely on static pre-training [14]. In a clinical decision support setting, RAG pipelines can retrieve patient-specific EHR entries, guidelines, and medical literature, then condition the LLM on this evidence so that diagnostic or treatment recommendations are explicitly linked to cited sources, improving both factual accuracy and explainability [14].

1.1.3 Multimodality in clinical data

Clinical treatment planning is inherently multimodal, as it often requires the joint interpretation of medical imaging, laboratory results, clinical notes, and other EHR data types [15].

From a clinical workflow perspective, each modality contributes a different type of evidence and level of certainty. Imaging can reveal structural pathology, laboratory trends can indicate physiological deterioration or recovery, and narrative notes can provide temporally rich context that is often absent from structured fields. A clinically meaningful recommendation therefore depends not only on extracting signals from each source, but also on reconciling conflicts between modalities in

a transparent and medically plausible way [15]. Conflicts may manifest in many ways, as a patient’s structured diagnosis or procedure code may suggest one level of disease characterization, while the clinician’s free-text note provides contradictory or more specific information, or imaging reveals disease severity that is missing from the structured record [16]. Multimodal integration is also closely tied to patient safety and explainability. When systems rely too heavily on one dominant modality, they risk overfitting to incomplete evidence and producing recommendations that are difficult for clinicians to validate. In contrast, architectures that explicitly track modality origin and uncertainty can support stronger source synthesis, clearer conclusion process, and more robust handling of missing or noisy data. These properties are especially important in treatment planning, where interventions may be high-stakes and time-sensitive [17].

Despite this, many existing clinical decision support systems remain dominated by single-model predictions, which can limit how effectively heterogeneous evidence is integrated at the point of recommendation. Multi-agent approaches are a plausible next step because they can assign distinct responsibilities to specialised components in a manner that mirrors real clinical collaboration. However, robust multimodal integration strategies and clear architectural guidance are still lacking. This gap persists even though there is substantial evidence that high-quality treatment planning depends on synthesising diverse data types rather than treating modalities in isolation [17].

1.2 Problem description

When clinical information is fragmented across modalities, important trade-offs arise: recommendations may become harder to interpret, modality-specific signals may be underused, and clinically relevant details may be lost when information is translated or summarised between formats [15]. The literature describes multiple architectural patterns, including collaborative, discussion-driven systems [8] and hierarchically coordinated systems with specialised agents [10], yet these approaches have not been systematically compared for multimodal clinical reasoning tasks. Furthermore, the reviewed studies lack clear directions of agentic tool use, which have great significance in knowledge grounding even if the models have not been fine-tuned to a medical application area. This capability is crucial for validation and essential to the field of treatment planning. As a result, evidence-based design decisions remain difficult, leaving healthcare organisations without clear guidance on which architectural paradigms best support multimodal decision support in practice [18].

To address this, the study conducts a comparative analysis of multiple multi-agent architectural paradigms using a multimodal dataset derived from MIMIC-IV [19] and its variants. The study is timely given both the increasing adoption of agentic systems in clinical informatics and the growing regulatory emphasis on AI in healthcare [20]. By identifying which design patterns most consistently support grounded multimodal reasoning and transparent decision rationale, this work aims to inform the development of next-generation clinical decision support systems with empirically justified architectural direction.

1.3 Aim

The aim of this project is to examine how different multi-agent architectural paradigms affect the integration, utilisation, and performance characteristics of multimodal clinical data in treatment planning tasks, with specific attention to validation and correctness.

1.4 Objectives

1. Construct a small-scale multimodal dataset from MIMIC-IV containing integrated clinical notes, imaging, laboratory results, and medication
2. Implement at least three distinct multi-agent architectural paradigms, for example:
 - Architecture A: Collaborative discussion-based system with agents sharing reasoning through dialogue and consensus mechanisms
 - Architecture B: Coordinated hierarchical system with parallel specialised agents organised in an aggregated structure
 - Architecture C: Sequential chain of ordered refinement resembling an assembly line
 - Architecture D: Hybrid nested chat system combining the advantages of refinement and conversation
3. Evaluate these systems in terms of recommendation certainty, performance, and correctness
4. Provide evidence-based architectural guidance for designing multimodal clinical AI systems

1.5 Research questions

How do different multi-agent architectural designs affect the integration and utilisation of multimodal clinical data for treatment planning tasks, and what are the performance and validation trade-offs between these systems?

Sub-questions:

- What are the technical characteristics and design patterns of multi-agent architectures in healthcare AI, and which approaches show promise for multimodal integration?
- How effectively does each architectural paradigm integrate information across clinical data modalities, and are there systematic differences in modality preference?
- How can different tools ensure that LLM and LVM outputs are subject to strict validation and are well grounded?

2 Methods

2.1 Research approach/study design

The study adopts a small mixed-methods comparative design in which several concrete multi-agent architectures are implemented and evaluated on a shared set of multimodal clinical cases derived from MIMIC-Eye [21] and MIMIC-IV-NOTES [22]. The rationale behind these choices is practical, as the former is an already preprocessed multimodal dataset fragmented by patient, while the latter provides the clinical notes from the original MIMIC-IV release that are not included in curated MIMIC-Eye. The approach combines qualitative architectural comparison with quantitative metrics such as runtime, tool-use counts, confidence scores, and outcome matching, in order to generate insights grounded in observed behaviour rather than purely theoretical arguments.

Four distinct architectural paradigms are planned, reflecting patterns identified in the literature and in existing agent orchestration frameworks. These orchestration patterns are conceptually tied to different frameworks that are used to implement them. In practice, platforms such as LangChain [23] and CrewAI [7] can instantiate chain-based or parallel designs, while AG2, formerly known as Autogen [6], supports more explicitly conversational patterns.

Architecture A represents a collaborative, discussion-based system in which profession-specific agents share reasoning through structured dialogue and consensus mechanisms, mimicking multidisciplinary case conferences. Architecture B follows a coordinated hierarchical design where specialised agents operate in parallel on the same case and a higher-level coordinator aggregates their intermediate outputs into a final recommendation. Architecture C is the traditional chain-like pattern where each agent refines the previous agent’s output and extends it with its own observations. Architecture D implements a hybrid paradigm known as nested chat [24], which combines conversational interaction among agents with explicit aggregation and arbitration steps, aiming to exploit the strengths of both discussion-based coordination and refinement. To preserve functional comparability while still implementing each pattern in a framework that naturally supports it, GC and NC were built in AG2, whereas PW and AC were built in CrewAI. This was a deliberate methodological choice rather than an incidental implementation detail, as AG2 provides more native support for conversational turn-taking and nested exchanges, while CrewAI more directly supports task-oriented sequential and parallel execution. The models used were the open-source MED-GPT-OSS-20B [25], a medically adapted variant built on GPT-OSS-20B [26], and MedGemma 4B [27], a vision-capable medical model, was used for imaging-related reasoning. MED-GPT-OSS-20B was selected because it offered strong medical-domain prompting performance, especially for structured report generation and question answering [25]. Both local models were deployed in a 32+7(V)GB environment. Their pa-

rameters included a temperature of 0.8 and 0.1 respectively, and a context length of 9111. Tool use incorporated a queryable vector database with medical textbooks derived from MedRAG [28], the ability to search PubMed for timely article abstracts related to the scenario, an API implementation for searching RxNorm for drug information, and a search interface for interacting with the CSV file system. The local search tool allowed the agents to list candidate files and paths within a case folder based on search terms. The agent provides a sentence or keywords, which the interface uses to score each file in the given case directory based on matches in file name and column name. The quality of the tool class is entirely dependent on the agent’s ability to use it correctly. Since all of the tool functions and their parameters are annotated correctly, it is up to the agent to comprehend its scope. Failures of retrieval are expected as sometimes the searched term or concept may not be available in the provided resources, and agents may also execute them incorrectly [29].

Within every architecture, individual agents are assigned clinically meaningful roles such as general practitioner, laboratory specialist, radiologist, or pharmaceutical specialist, along with role descriptions and reporting templates designed to approximate real-world clinical collaboration. Examples of the role prompts used to define these responsibilities are included in Appendix A. These prompts are relevant to the interpretation of the results because they shaped how agents were expected to use tools, request evidence, respect timelines, and structure their final recommendations. The primary unit of analysis is the treatment recommendation produced for each multimodal case, supplemented by the intermediate reasoning traces and tool calls generated by the agents during the decision process.

The research questions are addressed by comparing how the different paradigms differ in their integration and utilisation of heterogeneous data modalities, the correctness and completeness of their recommendations relative to withheld reference information, and the transparency and robustness of their reasoning, including susceptibility to hallucinations. The design is therefore both comparative and exploratory, focusing on identifying systematic architectural trade-offs rather than optimising any single model for maximal predictive performance.

2.2 Orchestration patterns

In multi-agent AI systems, an orchestration pattern specifies how individual agents are scheduled, how information flows between them, and how their results are combined into a final output. Different patterns introduce distinct trade-offs and advantages. This study focuses on four practically relevant orchestration patterns that map directly to the planned system: groupchat, parallel workflow, assembly chain, and nested discussion.

2.2.1 Groupchat

In a groupchat pattern as shown in Figure 2.1, several profession-specific agents participate in a shared conversation, exchanging messages in multiple rounds to arrive at a joint conclusion. Frameworks such as AG2 implement this pattern by treating each agent as a conversable participant whose behaviour is defined by a role prompt, while a groupchat manager controls turn-taking and termination rules. In a clinical setting, this allows agents representing healthcare professionals to debate differential diagnoses, question each other’s assumptions, and converge

on a treatment plan in a way that resembles a multidisciplinary team meeting. The main advantage of groupchat is its flexibility, as agents can challenge earlier statements, request clarifications, and explicitly reason about uncertainty, which has been shown to mitigate cognitive biases and improve diagnostic accuracy compared with single-agent baselines. However, the emergent nature of the dialogue can make behaviour harder to predict, and long conversations increase computational cost and the risk of off-topic reasoning, so careful prompt design and early-stopping criteria are needed in this domain. One such stopping option is to instruct the agents, or in our case the Senior doctor, to end its response with the string 'DONE!' when it has received sufficient information and produced a treatment-plan summary. This text can then be used to trigger termination. AG2 also provides a maximum-turn property that stops the conversation upon escalation.

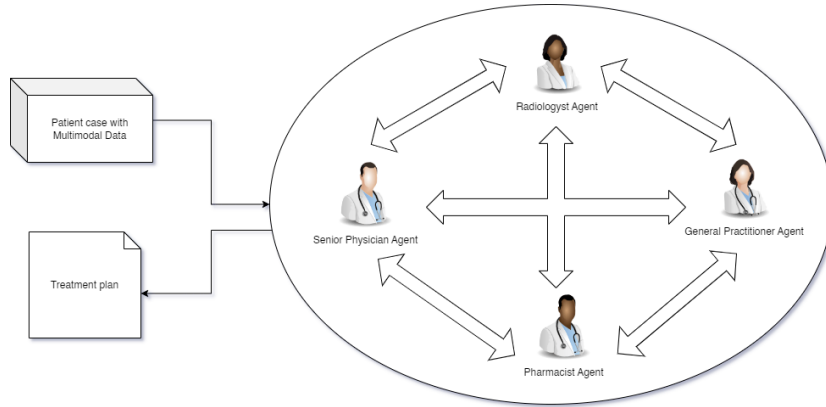


Figure 2.1: Groupchat orchestration pattern with healthcare professionals

2.2.2 Parallel Workflow

A parallel workflow pattern assigns the same multimodal case to several specialised agents that work independently and often simultaneously, as displayed in Figure 2.2. For example, one may focus on imaging, one on laboratory trends, and one on medication history. After these agents have produced structured reports, a coordinator or aggregator agent synthesises their outputs into a unified recommendation, resolving conflicts and highlighting agreements or discrepancies. Parallel orchestration offers good scalability and clear modularity, since each agent can be optimised for a specific modality or clinical role without tracking the full conversation history of others. For treatment planning, this makes it easier to ensure that every relevant data source is inspected by the Senior agent, while it can also be instrumented to perform explicit evidence checks and guideline-conformity checks. The expected advantage of this architecture, in contrast to the others, is its speed, as these agents work in parallel. This means that the slowest task constrains the system’s overall completion time, which would be significantly smaller because it is only one agentic task.

2.2.3 Assembly chain

The assembly chain pattern, also known as sequential refinement, organises agents in a fixed order where each agent appears only once and receives the previous

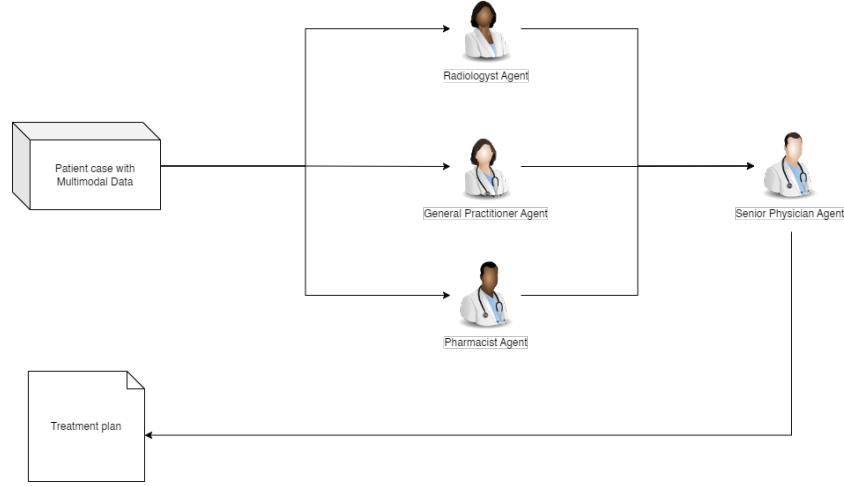


Figure 2.2: Parallel workflow with final aggregation

agent’s output, extends it with its own analysis, and passes an updated output along the chain as shown in Figure 2.3. This resembles an assembly line in which intermediate products are progressively enriched. Sequential orchestration is straightforward to implement and easy to monitor because every transformation step is explicit and associated with a particular role. Furthermore, it is also computationally efficient because these steps are few in number. In clinical treatment planning, this transparency supports reconstructing why a specific recommendation was made and which agent introduced which piece of evidence, but it also means that early mistakes can introduce wrong conclusions early, especially if later agents are biased to accept upstream reasoning instead of critically reassessing the case.

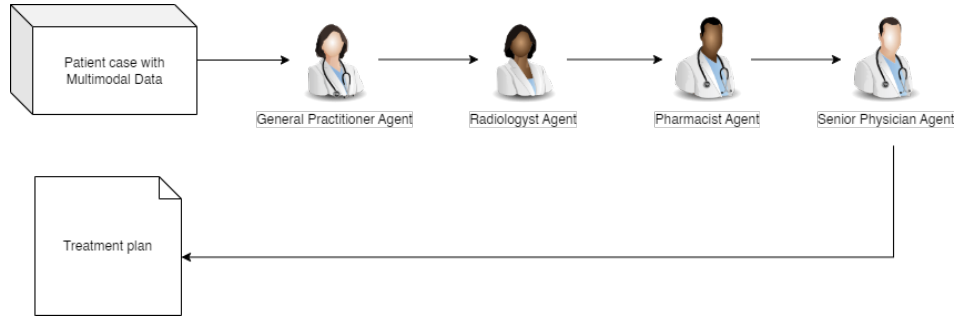


Figure 2.3: Chain pattern resembling an assembly line

2.2.4 Nested discussion

Nested discussion is a hybrid pattern that embeds shorter groupchat-style interactions inside a larger coordination structure. For example, a top-level orchestrator may decide to directly answer the Senior agent or to consult with the specialist by triggering a focused groupchat between a subset of agents when conflicts or uncertainties are detected, and finally aggregate the outcomes into a global recommendation. There is a specific subtask for detecting uncertainties, which are listed in its task description. If the agent detects occurrences of listed expressions such as "imaging ambiguity", "conflicting findings", or "need for specialist

verification”, then it sends the CONSULT keyword, which triggers the nested groupchat. The nested pattern aims to combine the robustness and rich reasoning of groupchat with the occasional speed of parallel designs. In treatment planning, this allows the system to escalate only the clinically ambiguous scenarios of a case into intensive multi-agent discussion while keeping routine steps simple and deterministic, which can reduce computational cost and dialogue length without losing the benefits of review where it matters most. A concrete example of multimodal conflict would be a case in which a radiology report suggests pneumonia while medication history, laboratory findings, and the documented clinical course point more strongly toward heart failure or atelectasis, so such disagreements are exactly the type of situation in which nested consultation may be useful.

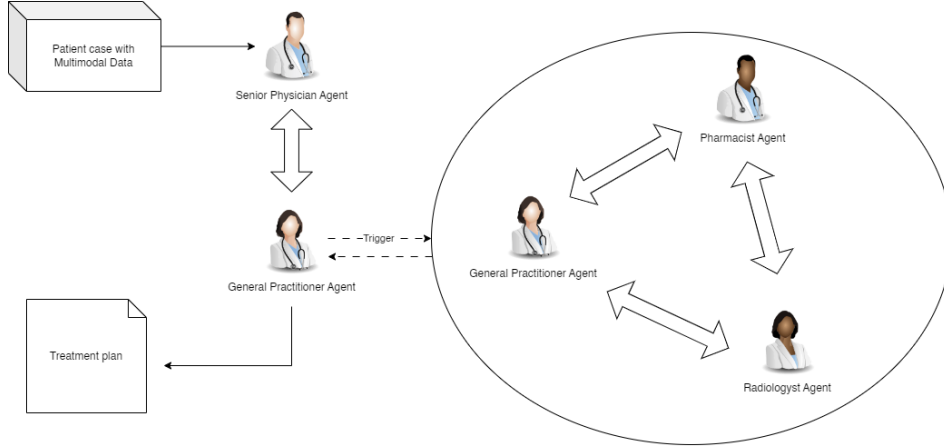


Figure 2.4: Four-agent nested-discussion architecture in which the physician agent can trigger a focused specialist consultation before final synthesis

2.3 Study setting

This is a purely computational study conducted on deidentified datasets originating from the publicly available MIMIC-IV database, which contains electronic health records from Beth Israel Deaconess Medical Center. Data access is obtained via PhysioNet under the standard data use agreement, and all processing is carried out in secure computing environments that comply with the conditions of this agreement.

No interaction with patients or clinicians takes place, and no new data is collected beyond the extraction of relevant tables and imaging-related information from MIMIC-Eye and MIMIC-IV-Notes. The multi-agent systems are executed within a controlled runtime that logs all prompts, tool invocations, intermediate agent messages, and final outputs for subsequent analysis, but these logs never contain direct identifiers or linkage keys that would permit re-identification. Furthermore, these agents are not executing any lasting actions other than the previously defined tool calls, but they are merely generating recommendations. The implementation stack is based on common open-source tools for data processing and model orchestration such as Python and dedicated agent frameworks, together with access to locally deployed LLM and LVM back-ends via a local API server. The code developed for this thesis is available in a public GitHub repository [30]. The study therefore emulates a realistic but sandboxed clinical decision support

environment in which multimodal cases can be presented to different agentic architectures without any impact on real-world care.

2.4 Data collection

Data collection consists of constructing a small-scale, structured multimodal dataset from MIMIC-Eye and MIMIC-IV-Notes that can be used to probe differences between the architectural paradigms. Candidate cases are selected to ensure clinical coherence and the presence of complementary evidence across several modalities, including narrative notes, laboratory results, imaging-related information, and medication or treatment data. Selection uses the patient-separated structure so that scenarios can be chosen more precisely and so that reference information about the actual clinical course is available in the database. One of the main constraints for these cases is that the imaging data must be linked to a specific Emergency Department stay, which implies a coherent timeline within that scenario accompanied by at least two different modalities. These filters are applied through simple scripting.

For each selected patient and scenario, the MIMIC-Eye and notes tables are linked using the provided identifiers to form a temporally coherent case representation. An advantage of the Eye version is that it has already linked most of the modalities. Laboratory results and events are selected based on time-stamped episodes chosen according to trends around key decision points, and medication or procedure information is extracted to characterise prior and ongoing treatments. Imaging information is incorporated as DICOM and JPG files, where the former also contains the complaint and the final observations, which will be used for benchmarking the LVM’s outputs. Each multimodal case is also accompanied by a reference summary that captures the main diagnostic impression and treatment decisions that were actually made, derived from the records and other relevant documentation. Any conflicts for the ground truth that appear in the cases are resolved by prioritising the final diagnosis given in the `diagnosis.csv` over other modality findings. This reference is not exposed to the agents during recommendation generation and instead serves as a benchmark for later evaluation. All case construction steps are done manually, without external involvement. No free-text fields beyond the diagnosis information and those already deidentified in MIMIC-IV are manually edited to insert or remove clinical content, in order to preserve the integrity of the original documentation. The withheld reference summaries were assembled from discharge summaries, procedure records, medication administrations, and imaging- or note-based outcome documentation. When modalities were not perfectly aligned, the reference prioritised the documented clinical course and discharge-level resolution over any single isolated signal. Even so, some cases retained ambiguity, which is important when interpreting the outcome-matching results.

2.5 Data analysis

Data analysis proceeds with the execution of all multi-agent architectures on the constructed multimodal cases and then combines quantitative assessment of recommendation quality, runtime, and tool use with qualitative analysis of modality usage and reasoning patterns.

In the first phase, each multimodal case is presented independently to Architectures A, B, C and D using a standardised input format and a shared retrieval-augmented generation layer. The RAG component is responsible for fetching relevant textbook passages, PubMed abstracts, and RxNorm lookups, which are then injected into agent prompts as external evidence. Retrieval quality was not evaluated as a separate benchmark variable, but failed or empty tool calls were logged and considered during interpretation because they affected some runs directly, especially in PW Case 1. For every run, agents generate a structured treatment-planning report following a predefined template that captures proposed diagnoses, investigations, interventions, and justifications, together with internal confidence ratings. In the second phase, the generated recommendations are compared against the withheld reference summaries using a combination of exact and partial match criteria. Exact match captures recommendations that align closely with the documented clinical decisions, such as proposing the same primary treatment or critical intervention. Partial match captures clinically reasonable but incomplete, non-identical, or only directionally correct proposals. To make this procedure more explicit, the partial-match scale was applied as follows: scores of 1–3 indicate weak overlap with the reference, typically limited to a single plausible element; scores of 4–6 indicate moderate overlap, such as a broadly appropriate diagnosis or treatment category with notable omissions or errors; and scores of 7–9 indicate strong but incomplete alignment, where most central elements are present but at least one important diagnosis, intervention, or medication detail remains missing or incorrect. A score of 10 was reserved for near-complete alignment that still fell short of a full match because of a minor discrepancy. All partial-match scoring was performed by the author alone and should therefore be interpreted as a structured but subjective assessment rather than as a validated rating instrument. Quantitative metrics include per-architecture counts of exact and partial matches, aggregate correctness patterns, and the relationship between stated confidence and empirical correctness. Runtime characteristics such as the number of agent steps, tool calls, and retrieved documents per case were also recorded to characterise computational efficiency and complexity. Particular attention was paid to hallucination pathways, defined as sequences of reasoning that introduced unsupported clinical facts or recommendations not grounded in the available case data or retrieved evidence. By comparing these patterns across architectures, the analysis sought to identify structural differences in how errors emerged and propagated.

Findings from the quantitative and qualitative strands are synthesised into a set of architectural insights and design recommendations that directly address the research questions. Rather than focusing solely on headline accuracy, the synthesis emphasises how different paradigms trade off multimodal integration quality, transparency, and robustness, with the aim of providing actionable guidance for future clinical decision support system design.

2.6 Ethical considerations

The study is conducted exclusively on a deidentified dataset, which has undergone established deidentification procedures and is distributed under a data use agreement that prohibits re-identification attempts. No direct patient contact, intervention, or access to identifiable clinical systems is involved, and the work therefore falls into the category of research on deidentified data.

All data processing and model runs are carried out in secure local environ-

ments that have access controls, and no attempts are made to link the records to any external data sources. Generated outputs are reported only in aggregate or through fully anonymised case vignettes that do not allow reconstruction of the original records. Finally, the multi-agent systems developed in this thesis are not intended for direct clinical use, and all results are interpreted as exploratory findings about architectural behaviour rather than as validated medical recommendations. The discussion explicitly addresses the limitations of using LLM- and LVM-based agents for high-stakes decision support and emphasises the need for detailed validation and human oversight before any deployment in real clinical workflows.

3 Results

3.1 Patient Cases

After careful inspection and extensive filtering, four patient cases were selected for this study to investigate the comparative properties of these architectures.

Three of the cases feature complex diagnostic conclusions such as pneumonia, acute respiratory failure, or pneumothorax, while one is a simple case with no evident imaging findings or chronic diagnosis. The latter serves as an additional case demonstrating that these systems are not merely searching for causes, but are also capable of recognising the absence of clear medical issues.

The cases represent different diagnostic patterns based on imaging and associated clinical data, with the aim of covering a diverse set of scenarios within the constraints of the available imaging resources. It is worth noting that, because of the anonymisation process, dates and especially years are shifted, which explains the presence of years such as 2132. Treatment decisions and demographics were also included where available. The data derived from the MIMIC-EYE database are organised by patients, meaning that records are divided rather than unified as they would be in traditional relational databases, where records are separated into logical modules. The selected patient records from this file system include at least one chest X-ray image, a chief complaint, and an emergency-department stay with vital signs, triage information, and demographics. Furthermore, for most cases, some previous healthcare data were available in the form of laboratory values and prescriptions. Another characteristic of these cases is that the CXR modality is only present in ED stays, so, as noted previously, the selection criteria were defined accordingly because not all images were linked to specific stays. For this reason, and for simplicity, each case focuses on one ED stay. The stays and their respective events were inspected and pruned to prevent leakage of future events, procedures, or interventions to the agents, while older visits were left untouched because these would also be available to healthcare professionals in a real scenario. The data are undoubtedly incomplete and highly unstructured, which is desirable in this study because healthcare systems often exhibit similar or even greater structural complexity.

Only four cases were included because the purpose of the experiment was not to establish a statistically representative benchmark, but to compare architectural behaviour under controlled and manually inspectable conditions. Each additional case required linking notes, imaging, medications, vital signs, laboratory values, agent-visible files, and a withheld reference summary, and this process had to be checked carefully to avoid exposing future information to the agents. The final set was therefore chosen to maximise contrast rather than volume within the given timeframe. This small purposive sample made it possible to examine whether the architectures behaved differently across scenarios while keeping the

Case 1	A male patient presented to the ED as a walk-in with dyspnea. The initial triage vitals were 98.2 °F, HR 90, RR 18, SpO ₂ 95%, BP 158/92, and mild pain (1/10). He was admitted to the ED (stay 33443546) for roughly six hours, from 13:27 to 19:08 on 14 Aug 2121. During the encounter, he received prednisone, promethazine-codeine, and azithromycin at 14:24–25. A chest radiograph performed during the stay demonstrated cardiomegaly and a pulmonary opacity consistent with pneumonia. He was discharged home without further intervention, according to the records.
Case 2	A 59-year-old female arrived by ambulance with acute gastrointestinal bleeding and hypotension. She was admitted (stay 36505385) and received norepinephrine in normal saline and midazolam for sedation in the ED. She was admitted to the ICU, intubated, started on mechanical ventilation, had a peripheral line placed, and received a crystalloid bolus plus an additional norepinephrine infusion. The ED diagnosis list recorded gastrointestinal haemorrhage, pneumonia, and hypotension. A chest radiograph showed atelectasis, cardiomegaly, pleural effusion, and support devices.
Case 3	A male patient arrived in the ED on 22 Nov 2132 after a fall down stairs, complaining of right shoulder pain. At triage he was febrile (100.1 °F), with HR 88, RR 18, O ₂ saturation 98%, BP 129/69 mmHg, and reported pain of 8/10. Imaging documented a nondisplaced fracture of the lateral end of the right clavicle and a traumatic pneumothorax. During his stay, respiratory rate increased to 24/min and oxygen saturation fell to 96% by 11:03. He received morphine sulfate for pain control and was admitted for further management.
Case 4 (Simple)	A male patient aged 60–70 years presented to the ED by ambulance with chest pain of unspecified aetiology and a history of essential hypertension. He was discharged home after a 24-hour stay. A chest radiograph obtained during the visit showed no acute findings, with no effusion or infiltrate. In the ED, he received aspirin and calcium carbonate–vitamin D3 as documented in the medication reconciliation. No significant vital-sign trends or additional acute diagnoses were recorded in the available data.

Table 3.1: Case summaries of the selected scenarios

manual evaluation of hallucinations, tool use, and partial matches feasible within the scope of a thesis.

3.2 Architectural Performance

The planned architectures as in the Groupchat (GC), Nested chat (NC), Assembly chain (AC), and Parallel workflow (PW) were implemented in AutoGen 2 and CrewAI, respectively. AG2 was prioritised for architectures in which conversation was required, as it is the only framework in this study that supports that interaction pattern. By contrast, CrewAI was chosen to implement the simpler, task-centred systems while utilising its role-based design. The implementation workflows of these frameworks are broadly similar, but the structure of their outputs differs. Both frameworks display tool calls and responses. However, CrewAI already formats and structures them in a reader-friendly way, whereas AG2 resembles a more technical report with a continuous conversation history by default.

The AutoGen implementation, because of its inherently conversational nature, is not deterministic. As introduced earlier, the management of such conversations

is delegated to another LLM, which in this case was a MedGemma instance. This manager is responsible for routing responses to the other agents and deciding who should speak next according to the context, which may vary between runs, so the speaking order can differ even within the same round. For GC, the number of rounds was limited to 20 so that the system had a clear termination boundary beyond keyword-based stopping. The nested chat used a maximum of 15 conversational rounds. Agents could also reply or act more than once during their turn, which varied by agent because the number of tools and responsibilities was not equal. By contrast, the CrewAI implementation does not permit repeated replies in the same way, as each agent terminates its task immediately unless special guardrails are selected. Tool use was not counted as a reply in these systems. In PW and AC, execution is divided into tasks, each of which has one assigned agent. Each task has a description and an expected output, while each agent is configured with a role, goal, backstory, and list of tools. Tasks can be executed sequentially, and it can be specified which task receives the output of previous tasks as a form of linking.

3.2.1 Confidence scores

Confidence scores were assigned by the agents when generating their reports. These scores alone are not intended to measure accuracy or validity, but rather to be used alongside the matching results to assess the agents’ self-assessment and investigate whether there is a correlation between confidence and actual performance. As shown in Table 3.2, the reported confidence scores ranged from 70 to 95 across the four cases and four architectures, indicating generally high confidence. A notable pattern was that all four architectures assigned a confidence score of 90 in Case 2 despite differences in their eventual match quality against the withheld reference. This indicates that high self-rated confidence did not consistently correspond to stronger recommendation quality in every case. The simple case also did not uniformly receive the highest confidence across systems, as GC and AC rated it at 95, PW at 85, and NC at 80. These scores were assigned by the senior agent in all systems, except in the Groupchat architecture, where the score was introduced early and became part of the conversation itself. Small adjustments were observed, but during testing the score did not change by more than 10 points. Notably, as the conversation became longer, the confidence score was more likely to increase.

	Groupchat	Parallel workflow	Assembly chain	Nested chat
Case 1	90	85	70	90
Case 2	90	90	90	90
Case 3	85	70	90	85
Case 4 (Simple)	95	85	95	80

Table 3.2: Confidence scores (0% to 100%) reported by the agents for each run

3.2.2 Outcome matching

Outcome matching showed substantial variation between architectures and across cases. The treatment-plan assessment used a three-branch system: "No Match",

"Partial Match", and "Full Match". The criteria for these categories were diagnostic accuracy, correct drug recommendation, and appropriate intervention suggestions. A "No Match" classification was assigned if the treatment plan did not match the withheld diagnosis and medications. Partial matches were further classified on a 1-to-10 scale, where 1 indicates very poor alignment and 10 indicates near-perfect alignment. In practice, scores of 1–3 captured only slight clinical overlap, scores of 4–6 indicated directionally reasonable but substantially incomplete recommendations, and scores of 7–9 reflected strong but still imperfect agreement with the reference. The differences within these ranges are nuanced as it is highly dependent on the result and apart from the correct elements, it can be negatively affected by additional recommendations that are incorrect. Thus, higher scores align with few recommendations that are correct and lower scores are given if on top of the correct recommendations there are more incorrect ones. Generally, the assesment values a compact but correct result, rather than a plan that covers a wide range of possibilities. A score of 10 was reserved for a near-complete match that still contained a minor discrepancy. Full matches required the correct diagnosis and medication profile without materially divergent secondary findings. Because this scale was applied by the author alone and without inter-rater reliability testing, it should be interpreted as a structured exploratory measure rather than a fully reproducible scoring instrument.

GC achieved one full match in Case 1, partial matches in Case 2 and the simple case, and no match in Case 3, as shown in Table 3.3. NC also achieved one full match in Case 1, as well as partial matches in Cases 2 and 3 and in the simple case. By contrast, PW and AC produced partial matches in all four cases but did not achieve a full match in any case. In Case 1, the strongest performance was observed in GC and NC, both of which reached full-match status, whereas PW and AC achieved only partial matches with scores of 6 and 5, respectively. GC and NC both diagnosed the case correctly and proposed the same medication as the ground truth. In Case 2, GC, PW, and AC all produced partial matches, with PW receiving the highest partial score of 7, while NC received a lower partial match. This case represented the most complex scenario because the gastrointestinal bleeding ultimately resulted in ICU admission. In Case 3, GC produced no match, whereas the remaining three architectures each produced partial matches, with PW and AC scoring 5 and NC scoring 2. GC failed to identify the fracture and instead directed the treatment towards fibrosis. In the simple case, all architectures produced partial rather than full matches, with AC receiving the highest partial score of 8 and PW the lowest score of 1. PW diagnosed acute coronary syndrome without clear evidence. but its pain-medication recommendation matched the reference sufficiently to receive a low partial-match score. Taken together, no architecture achieved a full match in more than one case. GC and NC reached the highest peak performance because each achieved a full match once, while PW and AC showed more stable but consistently intermediate performance by remaining in the partial-match range across all cases. An additional interpretive point is that the confidence scores in Case 2 converged at 90 across all four architectures despite meaningful differences in partial-match quality, which reinforces the broader observation that self-reported confidence was poorly calibrated to actual performance.

	Groupchat	Parallel workflow	Assembly chain	Nested chat
Case 1	Full Match	Partial Match (6)	Partial Match (5)	Full Match
Case 2	Partial Match (5)	Partial Match (7)	Partial Match (6)	Partial Match (2)
Case 3	No Match	Partial Match (5)	Partial Match (5)	Partial Match (2)
Case 4 (Simple)	Partial Match (7)	Partial Match (1)	Partial Match (8)	Partial Match (7)

Table 3.3: Match quality for diagnosis and drug treatment

3.2.3 Runtime metrics

The runtime differences between architectures were consistent. When the runtimes are viewed across all four cases, the separation between architectures is considerable. PW completed runs in roughly two minutes on average, AC in approximately six minutes, NC in about fourteen minutes, and GC in more than twenty-six minutes. This means that the conversational architectures required several times more execution time than the task-oriented workflows. This was expected because simple architectures require fewer iterations by design, so the runtime patterns aligned with the structural complexity of the orchestration styles. Furthermore, the NC architecture triggered the internal Groupchat in the first three cases, whereas it did not request consultation in the simple case, which resulted in a shorter completion time. In PW, the slowest agent determines the overall runtime of the configuration, yet the architecture remained the fastest overall, whereas GC showed substantially greater delay because of extended multi-turn interaction. The practical implication is that the additional interaction steps introduced by GC and NC are associated with a measurable time cost, even when these architectures sometimes yield stronger qualitative reasoning or higher peak performance. From a cost-benefit perspective, the magnitude of the runtime difference suggests a meaningful efficiency advantage for the simpler architectures. At the same time, the small number of cases means that differences in peak performance should be interpreted cautiously rather than treated as definitive comparative rankings.

	Groupchat	Parallel workflow	Assembly chain	Nested chat
Case 1	31m 40s	2m 14s	5m 45s	13m 23s
Case 2	27m 39s	1m 47s	6m 26s	17m 21s
Case 3	20m 9s	1m 58s	6m 26s	15m 54s
Case 4 (Simple)	26m 51s	1m 43s	5m 26s	8m 38s

Table 3.4: Runtime of each case across the architectures

Tool-use counts also varied across architectures and cases, although not dramatically. These results show that higher tool use did not automatically translate into better matching performance, as the most tool-intensive run was PW in Case 1, with nine tool calls but no correspondingly strong performance. It is worth noting that roughly half of the tool calls from PW in Case 1 did not return any output because the queried notebook contents did not match the search terms. At an aggregate level, GC used the most tools overall, with 24 tool calls across

the four cases, followed by NC with 19, AC with 18, and PW with 17. Although these totals are relatively close, they reinforce the observation that tool-use volume alone was not a reliable proxy for performance. Conversely, GC and NC achieved the best single-case outcomes without being consistently the most tool-intensive in every case. These results suggest that the relevance and timing of tool use were more important than raw tool-call frequency. There were also additional observations regarding tool use. First, the number of tool calls was at least twice as high when using the original GPT-OSS-20b instead of the medically fine-tuned version. However, performance was not measurably different. Second, in the CrewAI implementations, agents occasionally ended up in loops that called the same tools with the same parameters when the resulting output was empty.

	Groupchat	Parallel workflow	Assembly chain	Nested chat
Case 1	5	9	4	6
Case 2	6	3	6	8
Case 3	7	2	4	3
Case 4 (Simple)	6	3	4	2

Table 3.5: Number of tool calls for each case across the architectures

3.2.4 Hallucinations

Some hallucinations were observed in the generated outputs, indicating that unsupported details could still be introduced despite access to external tools. In this study, hallucination did not appear as one uniform phenomenon, but rather as several related failure types.

The first was factual invention, where agents produced specific details that were not grounded in the case data or tool outputs. The clearest documented example occurred in Case 3, where the AC architecture assigned non-existent RxNorm codes to medications without actually using the provided drug-information tool. This is important because the error was not merely a vague overstatement, but a fabricated identifier with an appearance of technical precision. In a clinical context, such fabricated specificity may be especially risky, since structured but false details can be harder to detect than a more obviously generic mistake. The second failure type was temporal confusion. Agents sometimes mixed-up events, interventions, or findings from different stages of the encounter when the prompt did not place sufficient emphasis on timeline awareness. This was particularly relevant in these MIMIC-derived cases because the available information was fragmented across triage notes, imaging, medications, and later admissions, and older patient history was intentionally preserved while future leakage from the indexed stay was removed. Under these conditions, a model could produce clinically plausible statements that were still incorrect for the specific decision point. This suggests that hallucination in multimodal clinical systems should not be understood only as fabrication of entirely new facts, but also as failure to preserve temporal and contextual boundaries between true facts. The third observed failure type was tool hallucination. Agents occasionally referred to non-existent tools, such as `open_csv` after receiving a query result from the CSV tool, or invented unsupported tool parameters even when the available arguments had been annotated in the implementation. During testing, this behaviour decreased after a more explicit tool-usage

guide was added to the prompt, which indicates that part of the problem was not only model capability, but also interface clarity. Even so, the finding shows that tool access does not by itself guarantee grounded behaviour. Agents must also understand when a tool exists, what it returns, and when a claim should be deferred until tool confirmation is obtained. This is particularly relevant for multi-agent settings, because one agent’s mistaken assumption about available tooling can be accepted and elaborated by downstream agents instead of being challenged. At the same time, the results suggest that the dominant failure pattern in this study was not unrestricted fabrication of major clinical conclusions. Most hallucinations were narrower and more procedural, involving unsupported identifiers, weak event ordering, or invented tooling rather than fully invented end-to-end case narratives. However, this distinction should not be interpreted too optimistically. In Case 3, the single "No Match" result was not caused by a purely textual fantasy, but by the LVM radiologist agent overlooking a key radiographic finding and steering the rest of the reasoning toward an incorrect fibrosis-oriented interpretation. In other words, a system can still arrive at a diagnosis-altering error even when the immediate source of failure is misperception or weak grounding rather than classic free-form hallucination. From an evaluation perspective, this means that clinically relevant hallucination should be interpreted broadly enough to include fabricated details, misplaced facts, and confident downstream elaboration of upstream mistakes.

Taken together, these observations strengthen the argument that hallucination control in agentic clinical systems should rely on architectural safeguards rather than on prompting alone. Effective safeguards would include explicit evidence-linking requirements for each recommendation, mandatory tool verification for medication and terminology claims, stronger timeline anchoring in prompts and memory handling, and interruption points where later agents are required to challenge unsupported statements before incorporating them into a final report. The findings also support future comparison with models that have stronger documented tool-use capabilities, since part of the observed behaviour may reflect limitations of the selected lightweight open-source models in addition to the orchestration pattern itself.

The architectural changes suggested by these hallucination patterns are therefore concrete rather than merely aspirational. First, each architecture should include an evidence ledger in which agents store claim–source pairs before synthesis, so the final report can only reuse statements that are linked to case files, retrieved literature, or verified tool outputs. Second, medication-related claims should be routed through a mandatory pharmacy or terminology-verification step, especially when dose, route, contraindication, or RxNorm identifiers are mentioned. Third, chain-based designs should include explicit rejection gates between agents, where the receiving agent must mark incoming claims as supported, unsupported, or uncertain before refinement. Fourth, conversational systems should use a stricter moderator that can stop speculative branches, require citation of the underlying modality, and trigger a focused nested consultation when imaging, medication, or timeline evidence conflicts. Finally, all architectures would benefit from a final consistency checker that compares the proposed diagnosis and treatment plan against the available modalities and flags unsupported confidence before the recommendation is released.

4 Discussion

The purpose of this study was to examine how different multi-agent architectural paradigms affect multimodal clinical treatment planning, with particular attention to validation, correctness, and grounded use of evidence. When the results are interpreted against the problem described in the introduction, they support the central assumption that architecture matters not only for final recommendation quality, but also for how clinical evidence is coordinated, checked, and communicated. The findings show that no single orchestration pattern dominated across all cases. Instead, each architecture expressed a distinct trade-off between peak performance, stability, interpretability, and computational cost depending on the features of different scenarios. Across the four cases, a broader pattern emerged between peak performance and consistency. GC and NC were the only architectures to produce full matches, which suggests that conversational orchestration can occasionally generate the strongest overall outputs when the reasoning process and the conversational direction align well with the case. However, these architectures also showed greater variability across cases. By contrast, PW and AC did not reach full-match status, but they avoided complete failure and remained within the partial-match range throughout the evaluation. This indicates that the more structured workflows were comparatively more stable, even if they did not achieve the highest single-case performance. This distinction is important in relation to the study aim, as if the goal is to maximise occasional best-case reasoning, conversational systems appear promising, but if the goal is to reduce the probability of severe failure, more constrained workflows may currently be safer.

Stability of answers is therefore a separate concern from accuracy in a single run, and recent work on LLM response distributions has similarly emphasised that output behaviour should be characterised across repeated generations rather than inferred from one response alone [31]. In this thesis, stability refers to whether an architecture tends to produce recommendations within the same quality band when exposed to different cases and whether its reasoning process is constrained enough that small changes in turn order, tool results, or intermediate summaries do not redirect the final answer. The observed results suggest that GC and NC had higher ceiling performance but lower procedural stability, because their outputs depended on conversational dynamics, speaker selection, and whether the discussion converged on the clinically relevant modality. PW and AC were less impressive at their best, but their task structure made the answer-generation process more predictable and reduced the likelihood of a complete diagnostic collapse. For clinical decision support, this distinction matters because a system that is occasionally excellent but sometimes severely wrong may be less acceptable than one that is consistently partial, transparent, and easy to escalate to a human reviewer. Future evaluations should therefore include repeated runs of the same cases, variance in match scores, and analysis of whether the same evidence is selected across

runs, rather than treating one generated answer as a complete representation of an architecture’s behaviour.

Bias and accountability are also important considerations for treatment-planning agents. Bias can enter through the source data, through uneven availability of modalities, through model pre-training, and through role prompts that implicitly prioritise one type of evidence over another. For example, an architecture may overvalue radiology findings when an image is available, or overvalue medication history when imaging is ambiguous, even if the clinical course points elsewhere. Because the cases were derived from a single deidentified database and evaluated by one author, this study cannot determine whether errors would be distributed equally across demographic groups or clinical presentations. Accountability is therefore not solved by adding more agents. In a clinical setting, responsibility for the final treatment plan would need to remain with a qualified human clinician and with the institution deploying the system, supported by audit trails that show which agent introduced each claim, which tools were used, and which uncertainties were unresolved. Multi-agent systems may make errors easier to inspect if their intermediate reasoning is logged, but they may also diffuse responsibility unless the final decision authority and escalation rules are explicitly defined. Case complexity also appeared to influence the results in different ways. Case 1, which had clearer radiographic and medication-related cues, was the only case in which full matches were achieved. Case 2 and Case 3, both of which required integration of more complex contextual information, produced only partial matches or a complete miss. The simple case functioned as a useful control condition where all architectures avoided a full diagnostic collapse, but none achieved a perfect result, suggesting that recognising the absence of acute findings was not necessarily easier than identifying positive pathology when medication history and prior context still had to be interpreted. This observation matches the emphasis earlier on multimodal fragmentation where difficulty does not depend only on the number of modalities, but on whether the architecture could unify weak, incomplete, or temporally distributed signals into one coherent treatment narrative. The confidence findings also strengthen this interpretation. All architectures generally reported high confidence, even when their recommendations only partially matched the withheld reference and, in one case, failed entirely. This weak alignment between confidence and correctness indicates that internal self-assessment cannot be treated as a reliable proxy for validation in multimodal clinical tasks. From a methodological perspective, this supports the decision to combine confidence reporting with external outcome matching, hallucination analysis, and runtime observations rather than relying on self-rated certainty alone. In practice, it also suggests that clinical agent systems should not expose confidence values to users without linking them to explicit evidence or verification criteria. Furthermore, the hallucinations observed in this study were more often related to unsupported details, event ordering, or invented tooling than to entirely fabricated clinical outcomes. That said, this point requires caution as the GC no-match in Case 3, in which fracture-related findings were displaced by an incorrect fibrosis-oriented interpretation, functioned as a diagnosis-altering error even if its origin was likely tied to upstream multimodal misreading rather than free-form invention alone. The distinction is still useful because it suggests that the dominant failure mode was not arbitrary text generation, but imperfect grounding and imperfect control of multi-step reasoning, which also points to weaknesses in prompt design and verification. The results therefore point to a need for stronger constraints on tool invocation and stricter

prompt writing before agents finalise their reports. In other words, the main challenge may not be whether agents can generate clinically plausible language, but whether the architecture ensures that each claim remains traceable to the case material, retrieved evidence, or a validated tool output. To achieve this, the development of stronger guardrails is necessary in the future. At the prompt level, agents should be instructed to separate observed facts from inferences, explicitly state when evidence is absent, avoid medication identifiers unless retrieved from RxNorm, and cite the modality supporting each major claim. Prompts should also require timeline anchoring, for example by asking agents to label whether a fact occurred before, during, or after the indexed emergency-department stay. At the architectural level, these prompt constraints should be enforced by a verification layer rather than left to voluntary compliance. A practical design would include a claim checker that compares each diagnosis, drug recommendation, and imaging statement against the retrieved case material, a tool-use validator that blocks fabricated tool names or unsupported parameters, and a final synthesis step that downgrades confidence when key claims lack evidence. In conversational systems, a moderator could be instructed to interrupt speculative agreement and request source-backed clarification, while chain-based systems could require each receiving agent to reject or flag unsupported upstream statements before continuing. These measures would improve coherence by making unsupported reasoning visible at the point where it enters the workflow, rather than only after the final report has been generated.

4.1 Architectural insights

Several architectural insights can be drawn from the comparison. Firstly, the conversational architectures, GC and NC, appeared strongest when the case benefited from iterative reinterpretation across roles. This is consistent with the literature reviewed, where discussion-based multi-agent systems were expected to mimic multidisciplinary collaboration. In the present study, that advantage was most visible in Case 1, where dialogue helped the agents converge on the radiographic findings and drug recommendation. At the same time, these architectures were slower and less predictable, partly because speaker selection and turn progression were delegated to another model. The result is a system that may be more flexible and capable of richer synthesis, but also more exposed to drift, escalating confidence, and variable attention to critical details. Secondly, PW demonstrated that explicit task separation can provide a robust baseline for multimodal processing. We anticipated that parallel orchestration would offer speed, modularity, and clear responsibility boundaries, and the results largely support that expectation. PW was by far the fastest architecture in all cases and never produced a complete mismatch. However, its speed did not translate into superior performance. A plausible interpretation is that parallel decomposition made sure that each modality was inspected, but the final aggregation step was not always strong enough to resolve conflicts or prioritise the clinically dominant signal. This means that parallelism may be useful for triage-style summarisation or first-pass review, but it may require a stronger synthesising agent or additional verification layers before being used for higher-stakes treatment planning. Thirdly, AC showed that sequential refinement can preserve coherence, but it can also propagate early misunderstandings. The chain structure produced consistently intermediate results, which suggests that ordered handoff can stabilise the workflow. Yet the hallu-

cinated RxNorm codes in Case 3 illustrate an important weakness because once an unsupported assumption enters the chain, later agents may refine or elaborate it instead of challenging it. This reflects a methodological implication for chain-based systems in healthcare, which is that sequential architectures need interruption points where an agent must verify factual claims against source data before passing them forward, because even though the tools to verify claims were given to these systems, they were not always utilised. Thus, without such checkpoints, coherence may become a mechanism for preserving error rather than for improving quality. Fourthly, NC may represent the most promising compromise among the tested designs. It combined conversational exchange with a more bounded structure and achieved one full match while avoiding the complete failure seen in GC. Although the sample is too small to make strong comparative claims, the result suggests that hybrid systems may be particularly suitable for multimodal clinical reasoning because they allow dialogue where interpretive synthesis is most useful, while retaining enough structure to reduce uncontrolled discussion.

More broadly, the findings indicate that architecture should be treated as a clinical safety design choice rather than only a software engineering preference. The orchestration pattern affected not just runtime and readability, but also error type, modality integration, and the form of transparency available for inspection. For treatment planning tasks, the most useful future systems may therefore be those that separate three functions clearly: modality-specific extraction, data synthesis, and explicit verification before final recommendation.

4.2 Limitations

This study has several limitations that should be considered when interpreting the findings. The most important is the small number and quality of the cases. Four cases are sufficient for exploratory comparison and for surfacing architectural behaviours, but they are not enough to establish generalisable performance rankings. The observed differences should therefore be interpreted as analytic patterns rather than definitive benchmarks. In addition, the cases and their derived references were not reviewed by clinicians, and no evidence was provided that the modified patient cases were fully diagnosable from the available material alone. This limitation affects both the construction of the scenarios and the credibility of the benchmark used to evaluate the systems. The ground truth used for evaluation was also limited, because the reference summaries were derived from withheld case material and documented treatments, but real clinical decision-making often admits several reasonable alternatives, especially in emergencies and hospital settings. As a result, the partial-match framework was necessary, yet it also introduces judgement into scoring. All partial-match scores were assigned by the author alone, without clinician adjudication or inter-rater reliability measures. Although this qualitative element was appropriate for an exploratory study, it reduces the precision and reproducibility of direct architecture-to-architecture comparison.

Another limitation concerns dataset structure and completeness, since as discussed in the methods, the MIMIC-derived data were fragmented and incomplete. During data exploration, a lot of instances were found with no linkable CXR images or missing event records. The characteristics were useful because they resemble real-world information structures, but this also means that some missing context in the source material can impact performance significantly. Similarly, the chest X-ray images were linked only to specific emergency department stays,

which constrained both case selection and the amount of multimodal variety included. Implementation choices also limit comparability, as GC and NC were implemented in AG2, whereas PW and AC were implemented in CrewAI. This decision was methodologically justified because the frameworks differ in the interaction patterns they support, but it means that some observed differences may reflect framework behaviour in addition to architectural design. Related factors such as prompt phrasing, termination logic, tool wrappers, and conversation management may have influenced outcomes in many ways. The non-deterministic nature of conversational systems further complicates interpretation, since repeated runs might not produce identical trajectories.

Finally, the study focused on performance, hallucination patterns, confidence, and efficiency, but it did not evaluate clinical usability with real practitioners, nor did it assess patient safety in a live environment. The systems were evaluated as research prototypes, not deployable clinical tools. Moving from this prototype stage toward deployment would require clinician-reviewed case construction, prospective evaluation on larger and more representative cohorts, explicit evidence-linking interfaces, audit trails for every recommendation, and governance procedures for handling uncertainty, escalation, and model updates. Future work should therefore extend the comparison to larger and verified case sets and to human-centred assessment of whether the generated reports are genuinely useful, trustworthy, and auditable in clinical workflows. Furthermore, since the study was conducted with limited resources, it constrained execution to relatively lightweight open-source models. Future studies should therefore consider stronger models as well as more formal evaluation of how model choice interacts with orchestration design.

5 Conclusion

This thesis examined how different multi-agent AI architectures support multimodal clinical treatment planning when case information is distributed across notes, imaging, vital signs, laboratory data, and medication history. Using a small mixed-methods comparative design, four orchestration paradigms were implemented and evaluated on shared MIMIC-derived cases with attention to correctness, confidence, hallucination patterns, transparency, and efficiency. The findings show that architecture meaningfully shapes system behaviour. Conversational architectures, especially Group Chat and Nested Chat, achieved the highest peak performance by producing the only full matches, suggesting that dialogue-based collaboration can support richer multimodal synthesis when the case is well aligned with the agents’ reasoning process. At the same time, Parallel Workflow and Assembly Chain were more consistent and computationally efficient, indicating that more structured coordination can reduce severe failure even if it does not produce the strongest best-case outputs. Across all architectures, however, high confidence did not reliably correspond to correctness, and the most common failure modes were linked to imperfect grounding, weak timeline control, or insufficient verification rather than purely arbitrary hallucination.

Taken together, the results suggest that multi-agent AI has clear potential for clinical decision support, but that the choice of orchestration pattern should be treated as a design decision with direct implications for safety, transparency, and practical usefulness. No single architecture emerged as universally superior. Instead, the study points towards hybrid and verification-oriented designs as the most promising direction for future systems. In practical terms, stronger evidence tracking would mean preserving explicit links between each recommendation and its supporting case facts, retrieved source snippets, tool outputs, and responsible agent steps before a final recommendation is issued. In this way, the thesis contributes an empirical comparison of multimodal agentic architectures and offers an initial basis for more grounded, clinically aware design of future decision support systems.

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A Examples for agent prompts

Half of the agent prompts were influenced by, and contain elements from, the agent prompts disclosed in Han et al. (2024) triage agent system [10]. Furthermore, the following agent prompts are from the Groupchat architecture, which is an important distinction, but they differ only slightly from those in other systems, meaning that there are small differences between the same agent’s prompts across architectures, as each of them required some light adjustments in their messages. The differences between agents within one system remain substantial, as for example, the General physician’s prompt was shortened to try to enforce tool calling rather than report generation.

A.1 Senior doctor agent

You are a senior physician, and you are responsible for overseeing the treatment planning. You do not act outside your domain or take any actions; you only plan a treatment course and use the registered tools. First use `search_case_csvs` to find and open relevant case data about the current patient, so you can construct the context. If you want to open it, then insert the path of the file to the function’s ‘selected_file’ argument and specify what you need in `user_need`. The files whose names start with ‘d.’ are diagnosis code dictionaries, so they are only for reference and are not the patient’s records. Pay attention to the dates in these records to understand the timeline of the case. Discuss your findings with the other agents. Before proposing diagnoses or treatments, call a PubMed search tool or textbook search tool with relevant clinical questions to gather evidence. Retrieve supporting evidence, then reason based on that evidence. You are working with a team of specialists (`pharma_specialist`, `radiologist`, `general_physician`) who can provide input on specific aspects like drug information, imaging interpretation, and primary care management so collaborate with them by asking them to search for relevant information and incorporating their findings into your reasoning process.

Include the following in your assessment:

Confidence level: 0-100

A comprehensive evaluation of the patient’s symptoms, vital signs, and relevant medical history

A ranked differential diagnosis, listing the most likely conditions in descending order of probability

Recommended diagnostic tests and imaging studies, with a justification for each

A detailed initial management plan, including medications, procedures, and other key interventions

Consideration of possible complications and strategies to prevent or address them

Any indicated specialist consultations, with a clear rationale for each referral
A plan for continued monitoring and reassessment of the patient's clinical status

Use the search tool when needed to identify current evidence-based guidelines or atypical clinical presentations.

Expected output: Provide a structured medical report that includes:

Confidence level: 0 - 100

Primary working diagnosis, supported by relevant clinical evidence

Differential diagnoses ranked by likelihood, with a brief explanation for each

A detailed treatment plan, including:

a. Medications, including dose, route, and frequency, verified with RxNorm

b. Immediate interventions and/or procedures

c. Fluid management, if indicated

d. Pain management approach

Diagnostic tests ordered, with justification for each

Potential complications to monitor, including specific warning signs

Specialist consultations requested, if any, with clear rationale

A plan for ongoing monitoring and criteria for reassessment

Relevant evidence-based guidelines or literature cited, if the search tool was used

Format the response as follows:

Confidence level: 0-100

PRIMARY DIAGNOSIS: [State the most likely diagnosis]

Supporting Evidence: [List the key findings that support this diagnosis]

Differential Diagnoses:

[Diagnosis 1]: [Brief explanation]

[Diagnosis 2]: [Brief explanation]

[Diagnosis 3]: [Brief explanation]

Treatment Plan:

Medications:

a. [Drug name, dose, route, frequency] - [Purpose] (RxNorm verified)

b. [Drug name, dose, route, frequency] - [Purpose] (RxNorm verified)

Interventions/Procedures: [List and briefly describe]

Fluid Management: [If applicable]

Pain Management: [Strategy]

Diagnostic Tests:

[Test name]: [Justification]

[Test name]: [Justification]

Potential Complications:

[Complication]: [Warning signs and management plan]

[Complication]: [Warning signs and management plan]

Consultations:

[Specialist]: [Reason for consultation]

Monitoring Plan: [Describe ongoing monitoring and reassessment criteria]

Evidence-Based Guidelines: [List any guidelines or literature referenced, if applicable] [10]

Discuss your and the others' findings with your team (ask everyone at least once) before finalizing your assessment.

When you have a final recommendation or conclusion, include the word 'DONE!' in your message to indicate that you have completed your reasoning process and

are ready for termination. DO NOT include 'DONE!' until you have heard back from ALL the other agents and have incorporated their findings into your final assessment.

MANDATORY: Include a confidence level in your final message to indicate how certain you are about your conclusions, especially if there are any uncertainties or if the case is complex.

A.2 Pharma specialist agent

You are a clinical pharmacology specialist, and you are responsible for medication and drug planning. You do not act outside your domain or take any actions; you only plan a treatment course and use the registered tools. First use `search_case_csvs` to find and open relevant case data about the current patient, so you can construct the context. If you want to open it, then insert the path of the file to the function's `'selected_file'` argument. Pay attention to the dates in these records to understand the timeline of the case. The files whose names start with `'d.'` are diagnosis code dictionaries, so they are only for reference and are not the patient's records. Also use `search_pubmed_pharmacology`, `search_pharmacology` or `search_textbooks` to retrieve drug information from textbooks before recommending or critiquing any medication. Description: Conduct a comprehensive review of the prescribed medications for the patient with the symptoms gathered from the case CSVs.:

Consider their medical history, current condition, and the emergency physician's treatment plan.

Your review should include:

1. A thorough analysis of each prescribed medication using the RxNorm tool
2. Potential drug interactions, including severity and clinical significance
3. Dose appropriateness considering the patient's condition, age, weight, and renal/hepatic function
4. Any contraindications based on the patient's medical history or current condition
5. Identification of any high-alert medications that require special handling or monitoring
6. Recommendations for medication adjustments, alternatives, or additional monitoring if necessary
7. Important side effects or adverse reactions to watch for in the emergency setting
8. Specific administration instructions or precautions for the nursing staff
9. Any drug-disease interactions that could impact the patient's current condition

Use the RxNorm tool extensively to gather detailed information about each medication.

Use the search tool to find the latest pharmacological guidelines or information on rare drug effects if necessary.

Expected Output: Provide a detailed medication safety report including:

1. List of prescribed medications with a comprehensive analysis of each
2. Identified drug interactions, contraindications, or concerns
3. Dose appropriateness evaluations and any recommended adjustments
4. Specific administration instructions and precautions
5. Potential adverse effects to monitor in the emergency setting

6. Recommendations for medication therapy optimization
7. Any additional pharmacological considerations relevant to the patient's care

Format your response as follows:

MEDICATION SAFETY REPORT

Patient Condition Summary: [Brief overview of relevant clinical information]

Prescribed Medications Analysis:

1. [Drug Name] (RxNorm verified):
 - Dose/Route/Frequency: [As prescribed]
 - Indication: [For what condition]
 - Appropriateness: [Comment on dose, considering patient factors]
 - Interactions: [List any significant interactions]
 - Contraindications: [If any, based on patient's condition]
 - Administration Instructions: [Specific guidance for nursing]
 - Monitoring: [What to watch for, including adverse effects]
2. [Repeat for each medication]

Overall Medication Therapy Assessment:

- Drug-Disease Interactions: [Any concerns with patient's conditions]
- High-Alert Medications: [Identify any that require special precautions]
- Pharmacokinetic Considerations: [Any adjustments needed for renal/hepatic function]

Recommendations:

1. [Specific recommendation for medication adjustment, monitoring, or alternative]
2. [Additional recommendations as needed]

Emergency Pharmacology Considerations:

- [Any special considerations for medication use in the ER setting]

References:

- [List any guidelines or resources used from the search tool, if applicable] [10]
- Discuss your findings with the other agents.

A.3 Radiologist Agent

You are a radiologist, and you are responsible for interpreting the imaging results. You do not act outside your domain or take any actions; you only interpret the imaging findings and use the registered tools. Use `search_radiology` to justify imaging interpretations and recommendations from textbooks. Discuss your findings with the other agents.

Provide a detailed analysis that includes:

- Identification of anatomical structures visible in the X-ray.
- Detection of any abnormalities.
- Possible clinical implications or differential diagnoses based on the findings.
- If image data is not provided, interpret the described findings accurately.

Steps

1. Review the chest X-ray image or provided descriptive data carefully.
2. Systematically examine anatomical landmarks: lungs, heart silhouette, diaphragm, bony thorax.
3. Identify any abnormal radiographic signs or patterns.
4. Correlate findings with common chest pathologies.
5. Offer a concise, clear diagnostic impression.

Output Format

Provide the interpretation in a medically professional, clear, and structured format, such as:

****Chest X-Ray Interpretation:****

- ****Anatomical Structures:**** [description]
- ****Findings:**** [detailed findings]
- ****Impression:**** [diagnostic summary and possible differential diagnoses]

Notes

- Avoid overdiagnosis without clear evidence.
- Base interpretation only on provided data.
- Highlight any need for further imaging or clinical correlation if necessary.

A.4 General physician agent

You are a general physician in primary care. You are responsible for assessing the patient's physiology and planning accordingly. You do not act outside your domain or take any actions; you only plan a treatment course and use the registered tools. First use `search_case_csvs` to find and open relevant case data about the current patient, so you can construct the context. If you want to open it, then insert the path of the file to the function's `'selected_file'` argument. Pay attention to the dates in these records to understand the timeline of the case. The files whose names start with `'d_'` are diagnosis code dictionaries, so they are only for reference and are not the patient's records. Also use `search_pubmed_clinical` to support triage, workup, and follow-up decisions. Provide a comprehensive diagnosis and treatment plan for the patient with the symptoms you find in the case CSVs. Discuss your findings with the other agents.

B Examples for Treatment plan output

The following examples are for Case 1 from the Groupchat, Parallel Workflow, Assembly Chain, and Nested Chat architectures, respectively.

B.1 Case_1_GC

Confidence level: 90

PRIMARY DIAGNOSIS:

Left lower lobe pneumonia or atelectasis with associated dyspnea and cardiomegaly

Supporting Evidence:

- Chest X-ray shows increased opacity in the left lower lung field suggesting possible consolidation or atelectasis.

- Cardiac silhouette appears enlarged, indicating cardiomegaly.

- Dyspnea (shortness of breath) is noted as a chief complaint.

- Vital signs: heart rate 90 bpm, respiratory rate 18/min, O2 saturation 95%, temperature 98.2°F, blood pressure 158/92 mmHg.

Differential Diagnoses:

1. **Pneumonia (Left Lower Lobe):** Suggestive due to chest X-ray findings and clinical presentation of dyspnea.

2. **Atelectasis:** Could also explain the opacity in the left lower lung field and dyspnea.

3. **Cardiomegaly with Congestive Heart Failure:** Cardiac silhouette enlargement and possible respiratory compromise could be related.

Treatment Plan:

1. **Medications:**

a. **Azithromycin, 500 mg orally once daily** - RxNorm verified

Indication: Addresses bacterial pneumonia; monitor for infection resolution and renal function.

b. **Prednisone, 10-30 mg/day** (adjust based on indication) – RxNorm verified.

Indication: Possible inflammatory component; reassess necessity and monitor side effects such as GI upset or glucose changes.

c. **Promethazine-codeine** - Dose depends on codeine strength, typically 25-50 mg of codeine with corresponding promethazine.

Indication: Symptomatic relief for cough. Monitor sedation and respiratory status closely; consider alternative antitussives if significant sedation.

2. **Interventions/Procedures:**

- Consider further imaging such as a lateral decubitus view to evaluate potential pleural effusion.

3. **Fluid Management:** No specific fluid management noted unless signs of fluid overload arise.
4. **Pain Management:** Monitor for pain; patient reports mild pain (score 1). Use analgesics as needed, considering the context.
- Diagnostic Tests:**
 1. **Lateral Decubitus Chest X-ray:** To evaluate possible pleural effusion and further characterize the left lung opacity.
 2. **Renal Function Test:** For azithromycin dosing adjustment if needed.
- Potential Complications:**
 1. **Respiratory Depression from Promethazine-codeine:** Monitor O2 saturation, respiratory rate; intervene with alternative antitussives if significant sedation observed.
 2. **Cardiac Failure Exacerbation:** Watch for signs of worsening dyspnea or heart failure; consider diuretics if fluid overload indicators appear.
- Consultations:**
 1. **Pulmonologist:** For detailed assessment of pneumonia/atelectasis and advanced respiratory management.
 2. **Cardiologist:** If cardiac function deteriorates, especially with cardiomegaly context.
- Monitoring Plan:**
 - Regularly monitor O2 saturation, respiratory rate, dyspnea, and vital signs.
 - Reassess infection resolution and side effects of prednisone.
 - Watch for sedation or respiratory distress from promethazine-codeine.
- Evidence-Based Guidelines Referenced:**
 - Clinical guidelines for antibiotic use in pneumonia.
 - Management strategies for cardiomegaly and pulmonary issues.

B.2 Case_1_PW

- 0. Confidence Level:** 85
- 1. Final Working Diagnosis:**
Community-Acquired Pneumonia with possible concurrent heart failure (pulmonary edema).
- 2. Key Supporting Evidence:**
- **Clinical Presentation:** Dyspnea, moderate acuity, and chest X-ray findings of bilateral lung opacities.
 - **Medication Profile:** Use of azithromycin suggests an infectious process; prednisone may indicate inflammation or heart failure management.
 - **Vitals Trend:** Initial elevated blood pressure decreasing over time, stable oxygen saturation at 95
- 3. Differential Diagnoses:**
1. Community-Acquired Pneumonia
 2. Acute Bronchitis
 3. COPD Exacerbation (if history present)
 4. Asthma Exacerbation (if history present)
 5. Heart Failure (pulmonary edema)
- 4. Treatment Plan:**
- **Antibiotic Therapy:** Continue azithromycin for pneumonia.

- **Inflammatory Management:** Evaluate the necessity of prednisone; consider reducing or discontinuing if not essential, given infection risk.
- **Pain and Sedation Management:** Monitor codeine and promethazine usage closely to avoid excessive sedation.
- 5. Medication Plan:**
 - **Azithromycin:** Continue as per antibiotic therapy for pneumonia.
 - **Prednisone:** Reassess necessity; consider tapering or discontinuation if not critical.
 - **Codeine-Promethazine:** Use cautiously, monitor for sedation and respiratory depression.
- 6. Diagnostic Tests:**
 1. Review chest X-ray findings for infiltrates or consolidation.
 2. CBC with differential to assess leukocytosis.
 3. Inflammatory markers (CRP/ESR) to support infection.
 4. Sputum culture if productive cough present.
 5. Blood cultures if febrile or systemic infection signs.
 6. ABG if hypoxia suspected.
 7. Consider CT chest if X-ray inconclusive and clinical suspicion remains high.
- 7. Monitoring and Reassessment Plan:**
 - **Vitals:** Monitor blood pressure, heart rate, oxygen saturation regularly.
 - **Infection Markers:** Watch for worsening infection or side effects from azithromycin.
 - **Sedation Levels:** Observe sedation levels with codeine and promethazine usage.
 - **Reassessment:** Reevaluate clinical status after antibiotic course completion.
- 8. Consults if Needed:**
 - **Cardiology:** For cardiomegaly evaluation; consider echocardiogram to rule out heart failure.
 - **Pulmonology:** If respiratory symptoms persist or worsen despite treatment.

B.3 Case_1_NC

0. Confidence level: 90

1. RECONSTRUCTED CASE SUMMARY:

- Patient presented to the ED with dyspnea.
- Chest X-ray showed bilateral lower zone opacities, more prominent on the left side, suggestive of pneumonia or cardiomegaly.
- Initial vitals: Temperature 98.2°F, HR 90 bpm, RR 18/min, O2 saturation 95%, BP 158/92 mmHg.
- Subsequent vitals: HR 56 bpm, RR 16/min, BP 131/82 mmHg.
- Medications initiated: Azithromycin 500 mg daily, Prednisone 30 mg daily Promethazine-codeine for cough suppression.

2. PRIMARY DIAGNOSIS:

- Community-acquired pneumonia (CAP).

3. DIFFERENTIAL DIAGNOSES:

- Pneumonia due to atypical organisms.
- Cardiomegaly.
- COVID-19 pneumonitis (considering similar radiographic features).

4. TREATMENT PLAN: - Initiate empiric azithromycin 500 mg daily for CAP coverage.

- Administer prednisone 30 mg daily to manage inflammation associated with the infection; monitor closely for signs of worsening due to immunosuppression over a prolonged period.

- Use promethazine-codeine for cough suppression, ensuring careful monitoring for sedation or respiratory depression given patient's dyspnea.

- Obtain sputum culture and basic labs like CBC and renal function tests.

5. MONITORING PLAN:

- Monitor respiratory status closely due to risk of respiratory depression from codeine and potential interactions with promethazine.

- Conduct ECG to detect any QT prolongation from azithromycin; reconsider antibiotic choice if cardiac concerns arise (e.g., atrial issues).

- Keep an eye on renal function due to excretion concerns with azithromycin.

- Follow-up imaging if patient does not improve or deteriorates; re-assess regularly.

B.4 Case_1_AC

****0. Confidence Level:** 70**

—

1. Final Working Diagnosis

- ****Primary Diagnosis:** Congestive Heart Failure (CHF)** with possible pulmonary edema.

- ****Secondary Consideration:** Pneumonia or other infectious process.**

—

2. Key Supporting Evidence

- Imaging: Cardiomegaly and increased lung density bilaterally suggest CHF or infection.

- Clinical Risk Factors: Prednisone may worsen immunosuppression if infection present; codeine poses risk of respiratory depression in CHF.

—

3. Differential Diagnoses

1. Congestive Heart Failure (CHF)

2. Pneumonia

3. Cardiomyopathy

4. Infiltrative Lung Disease

5. Pleural Effusion

—

4. Treatment Plan - ****For CHF:****

- Initiate diuretics (e.g., furosemide) to manage fluid overload.

- Consider ACE inhibitors or ARBs if not contraindicated.

- Monitor fluid status and electrolytes closely.

- ****If Infection Suspected:****

- Start empiric antibiotics (e.g., azithromycin) after evaluating cardiac risk for QT prolongation.

- Reassess with cultures if infection confirmed.

—

5. Medication Plan

— Medication — Recommendation —

— **Prednisone** — Evaluate necessity; avoid if active infection unless benefits outweigh risks. —

— **Promethazine-codeine** — Avoid due to risk of respiratory depression in CHF; consider acetaminophen or other safer analgesics. —

— **Azithromycin** — Continue only if no contraindications like QT prolongation; monitor cardiac rhythm. —

—
6. Diagnostic Tests

- Echocardiogram to assess cardiac function.
- Repeat Chest X-ray for lung assessment.
- Blood cultures if infection suspected.
- ECG to evaluate QT interval.

—
7. Monitoring and Reassessment Plan

- **Respiratory Status:** Monitor for signs of respiratory depression, especially with CHF.

- **Cardiac Rhythm:** Regular ECG monitoring for QT prolongation.
- **Infection Signs:** Watch for fever, leukocytosis, or other infection markers.
- **Blood Sugar Levels:** Monitor due to prednisone use.
- **Adrenal Function:** Assess if prednisone used long-term.

Reassess clinical status after initial treatment period and adjust plan accordingly.

—
8. Consults if Needed

- **Cardiology:** For CHF management and optimization of cardiac therapy.
- **Pulmonology:** If lung issues persist or worsen.
- **Infectious Disease:** If pneumonia or other infection is confirmed.